

Quant Trading Project - Plain-English Explainer

How we tested a stock-picking strategy, what we found, what we're running now, and what happens next. Written to be understandable without finance jargon.

Part 1 - The Backtest: what we tried, what failed, and what finally worked

A 'backtest' means: take years of past market data, pretend we're trading through it day by day, and measure how the strategy would have done. The trap is 'cheating' - letting the model secretly see the future. Most of our early attempts did exactly that without us realizing.

The journey, attempt by attempt

Attempt	What we tried	What happened
Versions 1-11 (early)	Various machine-learning stock pickers	Looked AMAZING - huge returns. But it was 'data leakage': the model was secretly peeking at future prices. Fake success.
Version 12 (rebuild)	Rebuilt the test so it CANNOT peek at the future (purged, leak-checked)	Results got much smaller - because now they were honest. This was the real turning point.
+ Company financials	Added SEC fundamentals (earnings, value, quality)	A fair test PROVED it made predictions WORSE, not better. Removed it.
+ Insider trading data	Added company-insider buying (SEC Form 4)	Same story - it LOWERED real skill. Removed it.
+ Fancier models	xgboost, ensembles, model stacking	No improvement over the simple model. Complexity didn't help.
Final: minimal model	Just two honest signals (volatility + relative ranking), and sit in cash when the market is dangerous	The ONLY version with a small but statistically REAL edge.

Why the final backtest counts as a success (honestly)

- The signal is statistically REAL, not luck: information-coefficient 0.055 with a t-stat of 2.04 (above 2 = unlikely to be chance), across 32 separate out-of-sample test windows (2018-2026).
- It cuts the worst crash roughly in HALF: -19% vs the S&P 500's -34%.
- Honest caveat: it does NOT beat the S&P 500. Same return-per-risk (Sharpe 0.74 vs 0.74), and the S&P actually made more money overall. It's a SAFER ride, not a richer one.
- The real win: it beats the cheap 'low-volatility' ETFs you could just buy (USMV, SPLV) - so it earns its existence as a defensive strategy.

The numbers (final backtest, 2018-2026, after trading costs)

Measure (plain meaning)	Our Strategy	S&P 500	Who wins
Sharpe (return per unit of risk)	0.74	0.74	Tie
Worst drop / crash	-19.0%	-33.7%	Strategy (half the crash)
Yearly bumpiness (volatility)	12.7%	19.3%	Strategy (calmer)
Total return over the period	99%	170%	S&P (more money)
Yearly return (CAGR)	9.0%	13.2%	S&P

Versus the off-the-shelf low-vol ETFs (the real alternative to a 'safe' strategy):

Measure	Our Strategy	USMV	SPLV
Sharpe (higher better)	0.74	0.64	0.53
Worst drop (lower better)	-19%	-33%	-36%
Calmar (return vs worst drop)	0.47	0.28	0.21

Our strategy wins on every risk measure here - notably HALF the drawdown of the ETFs. That is the thing that justifies it.

Our strategy vs the market and vs cheap low-vol ETFs (2018–2026, after costs)

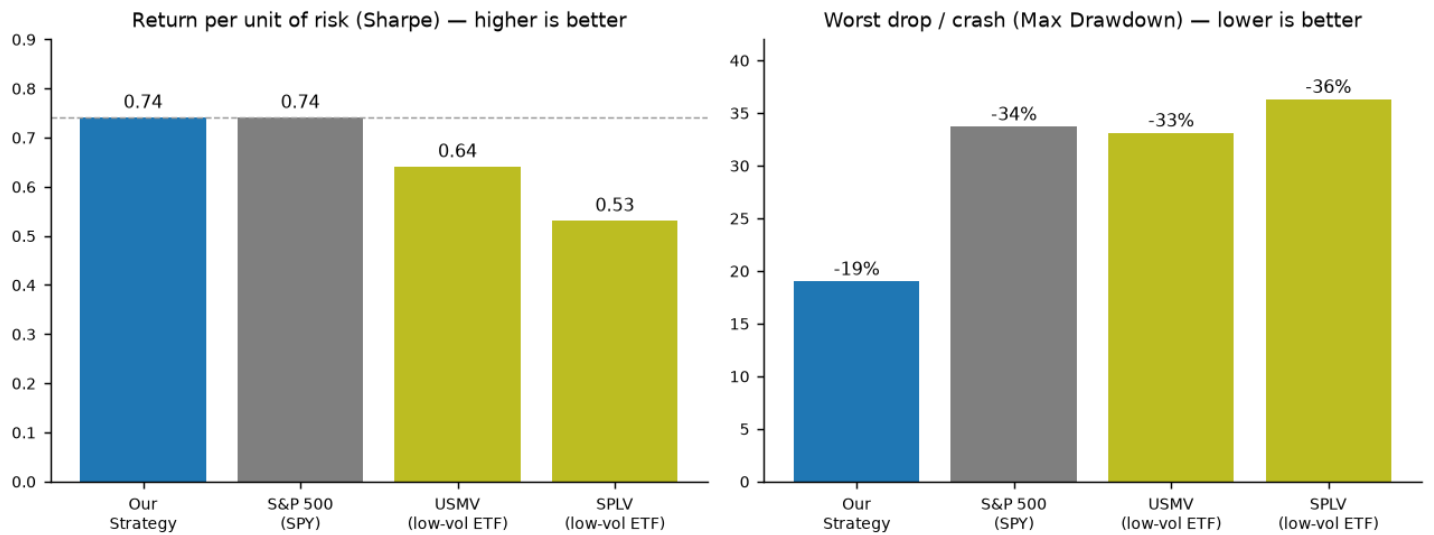


Chart: same return-per-risk as the market with a far smaller crash, and better than the cheap low-vol ETFs on both.

Part 2 - The Paper Test: what we're running right now

A backtest is the past. A 'paper test' is the FUTURE, traded with fake money in real time - the honest way to see if the edge is real going forward (and it can't be inflated by survivorship bias). Every weekday, automatically, the system makes real decisions and records them. Zero real money is at risk.

We run FOUR strategies side by side - a horse race - to answer one question: does all the extra machinery actually beat the simple validated strategy?

Horse (variant)	What it is
1. equity_validated	The simple validated strategy from Part 1. The 'control' / benchmark to beat.
2. equity_full_goal	Stocks, but with ALL the bells and whistles: 6 market 'regimes', position-size caps, Kelly risk sizing, correlation checks.
3. crypto_full_goal	The same full system, applied to crypto (Bitcoin, Ethereum + large coins).
4. full_system	Everything combined - stocks + crypto + a 'learning loop' that shifts money toward whichever is performing.

How it runs (hands-off)

- A robot job on GitHub runs every weekday after the US close - no computer of yours needs to be on.
- It logs each variant's decisions and its REAL next-day return into a running record (the 'ledger').
- It writes daily reports + a scoreboard (and now a CSV and PDF, optionally emailed to you).
- Each decision lists: the asset, buy/hold/sell, an expected-value score, a risk rating, confidence, position size, and the reasoning - full transparency.
- It needs about 20 trading days before the scoreboard is meaningful, and ~90 days for a real verdict.

Part 3 - What happens once the paper test is done (~3-6 months)

After ~90+ trading days of honest forward results, each variant faces a 'promotion gate'. It can graduate from fake money to real only if it clears ALL of these:

Gate check	Plain meaning
Positive real Sharpe	It actually made risk-adjusted money going forward, after costs.
Beats the simple baseline	The fancy machinery must out-perform variant 1 - or we keep the simple one.
Beats USMV / SPLV	It must still beat the cheap ETFs you could buy instead.
Respected the risk rules	No drops worse than the -8% stop allowed; went to cash in crises.

Then one of three honest outcomes:

- It clears the gate -> graduate to LIVE paper, then consider a SMALL amount of real capital, scaling only as it keeps proving itself.
- It only ties the simple baseline -> deploy just the simple validated strategy; drop the extra complexity.
- Nothing clears the gate -> the honest answer is 'no durable edge - just buy the ETF.' Concluding that truthfully is itself a success; it's what 11 earlier 'amazing' versions failed to admit.

New data sources (news, Reddit, options flow, etc.) only get added AFTER they pass the same evidence gate - so we never pay for or trust data that doesn't actually help.

Bottom line: we built a testing engine that refuses to lie to us, used it to find a small-but-real defensive edge that beats cheap alternatives, and we're now confirming it forward with fake money before risking a cent. Discipline over hype.